

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)
 Department of Computer Science and Engineering
ASSESSMENT TOOL 1 – SHORT ANSWER TEST (30 Questions)

Course Code:	ITA0506	Course Title:	COMPUTER VISION
CO Assessed:	CO1 – Imitation (BL3)	Assessment:	Assessment Tool 1 – Short Answer Test
Weightage:	40% of CIA	Total Marks:	50 (10 Questions × 5 Marks)
CO4 Statement:	Apply the fundamental concepts, image formation models, and processing techniques for analyzing Computer Vision systems. (BTL 3)		
Student Name:	K. Teja Sri	Reg. No.:	192525249
Section / Batch:	Slot D	Date:	11/7/26

Code of Conduct
 I K. Teja Sri (192525249) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: K. Teja Sri

Instructions

- This test contains 6 questions, each carrying 5 marks. Total: 30 Marks.
- No negative marking. Unanswered questions carry zero marks.
- No calculators or electronic devices permitted.
- Duration: 60 minutes.

Section / Topic	Focus	Q Nos	Marks
Overview of Computer Vision	Definition, objectives, workflow, and importance of Computer Vision	1	5
Features of Computer Vision	Feature extraction, object recognition, and image understanding	2	5
Levels of Computer Vision and Image Processing	Low-level, mid-level, and high-level image processing operations	3	5
Digital Image Fundamentals	Pixels, image representation, grayscale/color images, and image resolution	4,10	10
Image Sensing and Acquisition	Image sensors, acquisition process, and imaging devices	5	5
Sampling and Quantization	Sampling, quantization, and digital image formation	6	5
Image Formation Models	Perspective projection, illumination, and image formation process	7	5
Applications of Image Processing and Computer Vision	Healthcare, autonomous vehicles, surveillance, agriculture, robotics, and industrial automation	8,9	10
GRAND TOTAL:			50 Marks

Annexure A – Short Answer Test

1. How does Computer Vision enable computers to interpret and understand visual information?

- a) What are the main objectives of Computer Vision?
- b) Mention any two real-world applications of Computer Vision.

2. What are the key features of Computer Vision that make it suitable for intelligent image analysis?

- a) Define feature extraction.
- b) How does object recognition improve Computer Vision systems?

3. How are the different levels of image processing used in Computer Vision?

- a) What is low-level image processing?
- b) Differentiate between mid-level and high-level image processing.

4. How is a digital image represented and what are its fundamental components?

- a) Define a pixel.
- b) How does image resolution affect image quality?

5. How does image sensing and acquisition convert a real-world scene into a digital image?

- a) What is the role of an image sensor?
- b) Name any two image acquisition devices.

6. How do sampling and quantization contribute to the formation of a digital image?

- a) Define sampling.
- b) What is the purpose of quantization in image processing?

7. How does the image formation model describe the process of capturing images?

- a) What is perspective projection?
- b) What is the role of illumination in image formation?

8. How are image processing and Computer Vision related, and how do they differ?

- a) Define image processing.
- b) State two differences between image processing and Computer Vision.

9. How is Computer Vision applied in solving real-world problems across different domains?

- a) Explain one application in healthcare.
- b) Explain one application in autonomous vehicles.

10. How do digital image fundamentals support Computer Vision applications?

- a) Why are pixels considered the basic building blocks of digital images?
- b) How do sampling and image resolution influence image analysis?

Rubrics for Calculation

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning	Marks	
Understanding of Concepts	50%	Demonstrates thorough, accurate understanding of concepts	Good understanding with minor gaps	Basic understanding with some misconceptions	Poor understanding; major misconceptions		
Presentation	50%	Detailed, well-explained answers with relevant examples	Clear explanation with adequate details	Limited explanation, lacks depth	Poor or unclear explanation		

Faculty In-charge	Course Coordinator	Head of Department
Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____

1. How does - - - - -

Computer vision is a branch of Artificial Intelligence and It enables computer to capture, acquire, process, analyze and interpret the objects by seeing images and videos. is called computer vision. and it is used to make decisions.

a) objectives:

- 1) Image Enhancing and object recognition.
- 2) Understands the scenes
- 3) Image identification & object & shape recognition
- 4) Making the decisions by using interpret.

b) Two - real world

- 1) Face recognition
- 2) self-driving cars.

2. Key features of computer vision is detecting the images, recognition of objects, making decisions.

a) Feature extraction is the process of identifying the characteristics such as edges, corners, textures and shapes from an image is called Feature extraction.

b) object recognition identifies and classifies objects and recognize the data step by step & it makes a decision from an image for digital representation.

3. There are three-levels of image processing

- 1) low-level → Image Enhancing, noise removal
- 2) ~~High-level~~ medium-level → Image segmentation and Feature Extraction
- 3) High Level → making decisions and understanding scenes.

a) low-level:

low-level image processing is used for image enhancing & In this level it will remove noise & it is used to get good quality image.

b) mid-level	High-level
1) It Enhances Images 2) output is in quality-image.	1) It Understands Images 2) output is in information/decision.

4. Digital image represents the matrix pixel values. Each pixel values stored in intensity (or) colour in an image.

a) Define Pixel:

Pixel is a small amount of unit that stores the intensity (or) colour of the image from an Digital representation.

b) Image resolution is the process of identifying the quality of image if it is High pixel value it is High Quality if it is low it may affect to decrease the quality of image.

5. Image sensing is a process when the light is taken from the scenes and converts it into an electrical signals it is used for an digital image.

a) [The role of an image sensor is the light is converted into an electrical signal].

b) 1) digital camera.

2) web cam.

a) When the light is taken from an object and passes through camera lens through an digital image.

6. Sampling determines it converts continuous images into discrete pixels while quantization analyze the intensity values to that pixels.

a) sampling is a process of dividing convert continuous images into discrete pixels.

b) Quantization in image processing is used the intensity values into sample pixels for an image to get the digital representation.

7. Image formation model Explains how the light from the object Passes through an camera lens for capturing the image.

a) Perspective Projection is a process of Projecting an 3D scene to an 2D image Plane.

b) Illuminations Provides the light, required to capture, clear and quality image.

8. Image processing is used to improve the Image quality and computer vision is used to make the decisions by using internet.

a) Image processing is a technique of enhancement, restoring or manipulating the digital image.

b)

Image Processing	computer vision
1) Enhances images Output is improved quality	1) Understands images 2) Output is in information/decision.

9. Healthcare : computer vision detects the diseases such as X-rays, MRI scans, CT scans. It helps doctors in diagnosis.

Autonomous vehicles : It detects roads, traffic signals, pedestrian obstacles for safe navigation.

11.

Digital image fundamentals are used to detect the image & considering an object by the support of computer vision.

- a) Pixels are two types low level, ~~High level~~ so it is used to stores intensity and color in the images. so if it how it is how Quality, if it High it is High in quality based on the pixels.
- b) Sampling is used to count the no. of pixels and image resolution influences the images through recognition of analysis.